

General Active Global Stabilization of Multiloads DC-Power Networks

Pierre Magne, *Student Member, IEEE*, Babak Nahid-Mobarakeh, *Member, IEEE*, and Serge Pierfederici

Abstract—Implantation of complex dc-power network is one of the main research topics in more electric aircraft (MEA). In such applications, small and light systems are required and so optimization of passive elements, such as dc-bus capacitance and filtering inductance, is an important issue. It is known that the reduction of dc-bus capacitance may lead to instability of an MVdc network. So, if no stabilizer is used, the risk of instability must be considered, while designing the system passive elements. In this paper, we will first study the small signal stability of an MVdc network composed of three loads: an inverter supplying a permanent magnet synchronous motor, a dc/dc converter feeding a resistive load, and a supercapacitor controlled by a bidirectional dc/dc converter. Then, we will propose a large-signal-stabilizing system to ensure global stability by generating proper stabilizing power references for the whole system. The contribution of the loads to network stability is adjustable. The validity of the proposed method will be confirmed by experimentations.

Index Terms—AC drives, active stabilization, constant power load (CPL), dc-power network, energy storage device (ESD), fault-tolerant control, Lyapunov methods, more electric aircraft (MEA), MVdc networks, nonlinear systems, power electronics systems, power system stability, stability, supercapacitor (SC).

I. INTRODUCTION

AS The aircraft industry works on its next generation of engines, an important research and development topic is plane electrification. Significant progress in electrical system reliability, fault detection, and control in degraded mode have helped make “more electric aircraft” (MEA) more of a reality [1]–[4]. To optimize on-board energy management, the use of local dc-power networks with reversible loads and energy storage devices (ESDs) is a potential solution. The voltage range of these dc-power networks is situated between 200 and 600 V and such electric structures are often defined as MVdc in the literature. In this context, MVdc networks, including several actuators and ESDs, such as supercapacitors (SC) or batteries, are proposed and studied in the literature [5], [6].

In the aerospace industry, mass optimization is very important and designers are looking for smaller and lighter systems. As passive elements, dc-link capacitors and filter inductances, are

generally big and heavy, and so their optimization is a main research topic related to maximization of weight savings. Unfortunately, in the case of two cascaded systems, reducing the bus capacitor value decreases the system damping ratio and enhances instability risk due to the interaction between devices [7]–[11].

To study the instability phenomenon in dc-power networks, loads are often considered as “constant power load” (CPL), i.e., the impact of the dc-link voltage variations on the load power is negligible. This is almost true for tightly controlled converters while the load controllability conditions are satisfied. This approximation permits to consider a load in a restrictive case for stability problem and is applicable for all converters tightly regulated with a high bandwidth and good robustness properties as regard to possible disturbances. This is the case with inverter motor drive systems [8], [10], [12]–[14]. It is also partially true in a faulty mode, where the absorbed power has a significant dc component at high speeds [15].

Stability problems in the case of interaction between a power supply in cascade and a CPL is well known and numerous studies have been published to explain this phenomenon for the inverter motor drive systems [7]–[9]. In the case where the system is locally stable around an operating point, external disturbances like a change in the main voltage or an abrupt load power variation can make it unstable. To analyze this phenomenon, some authors have proposed mathematical methods to obtain an approximation of the domain of attraction of the operating point [16]–[20]. To prevent this instability risk, an appropriate design of passive elements could be done to obtain satisfactory stability margins. In [21], several passive methods were proposed and studied to damp the system and enhance its stability. Unfortunately, this also increases the global mass of the system.

To reduce the dc-link capacitor size and ensure a well-stabilized system, several stabilization methods have been proposed [10], [13], [14], [22]–[24]. They propose use of a feedback implementation on the load control improving the stability margins. This permits reduction of the dc-link capacitance. Experimental results prove the efficacy of these solutions but often the stability is studied and proved only for small signal variations around an operating point. In other works [25]–[28], the large signal stability of the system was considered and nonlinear stabilization techniques proposed.

In all these cases, stabilizing signals are superposed to the load power references to improve the system stability. This may be considered as a drawback in applications, where we cannot modify the loads dynamics. In [29], it was shown that it is theoretically possible to implement a proper feedback on a controllable voltage source to ensure the small signal stability of a power system composed by two cascaded converters.

Manuscript received December 21, 2010; revised July 19, 2011 and September 2, 2011; accepted September 3, 2011. Date of current version February 20, 2012. Recommended for publication by Associate Editor F. Blaabjerg.

The authors are with the Groupe de Recherche en Electronique et en Electrotechnique de Nancy (GREEN), INPL, University of Lorraine, Nancy 54516, France (e-mail: pierre.magne@ensem.inpl-nancy.fr; babak.nahidmobarakeh@ensem.inpl-nancy.fr; serge.pierfederici@ensem.inpl-nancy.fr).

Color versions of one or more of the figures in this paper are available online at <http://ieeexplore.ieee.org>.

Digital Object Identifier 10.1109/TPEL.2011.2168426

The stability is analyzed in the papers cited earlier only in the case of two cascaded systems. They have a good approach to the stability problem in dc-power systems, but they do not consider the case of a full dc-power network with several loads, which corresponds to the majority of industrial applications. To complete these studies, this paper deals with the stability problem in a multiloads dc-power network. A two-load MVdc system was studied in [30], but the authors used a local stabilizer for each load to ensure the small signal stability of the system by locally adjusting the input admittance of the loads.

The method proposed in this paper stabilizes a multiload dc-power network. Its main advantage lies in the mathematical proof of the global stability of the whole network. Nevertheless, to obtain this result, a stabilizing power should be absorbed by each CPL. To consider this aspect, which could be a negative in some applications, we propose to tune the stabilizer parameters in a way to share the stabilizing effort of each CPL. Most critical CPLs (actuators) have to make less effort than the least critical ones (ESD). It will be shown that with a proper mutual solicitation of each load, stabilizing powers are very small with respect to nominal powers. This will be validated by experimentations.

The proposed method is based, in a similar way than in [28], on the Lyapunov theory. In [28], for a switching converter, authors have proposed a class of control laws, which comes from a solution of the Lyapunov equation. Here, we propose a general way to design a stabilizing feedback, which also ensures that the nonlinear model of the system satisfies the Lyapunov equation. Nevertheless, the proposed approach is not limited to the case of a switching converter. It is applicable to every dc-power network with controllable loads. Indeed, its implementation is well adapted to dc-power networks as it permits to generate a n -dimensional stabilizing signal vector for a system with n CPLs. With this technique, the global stability of the system is achieved and the contribution of each CPL can be adjusted in a proper way. The use of a centralized stabilizer permits to control the impact of each CPL on the stabilization system.

This paper is organized as follows. The whole system and its control are presented in Section II. In Section III, a brief study of the system small signal stability around its operating point is presented. In Section IV, a method to stabilize the whole system by the generation of stabilizing powers for each CPL is discussed. Finally, in Section V, we confirm the efficacy of the proposed method by an experimental implantation and validation.

II. SYSTEM PRESENTATION

A. Electric Architecture

The proposed dc-power network consists of one dc voltage source and three different loads. The first load is an inverter supplying a permanent magnet synchronous motor (PMSM), the second is a power-controlled dc/dc converter that supplies a resistive load and the last one is an SC that is connected to the main dc-bus through a current-controlled bidirectional dc/dc converter. The loads are all different to show that the proposed method does not depend on the load type.

In an on-board MVdc network, and more particularly, in aeronautic applications, the voltage source cannot always be placed

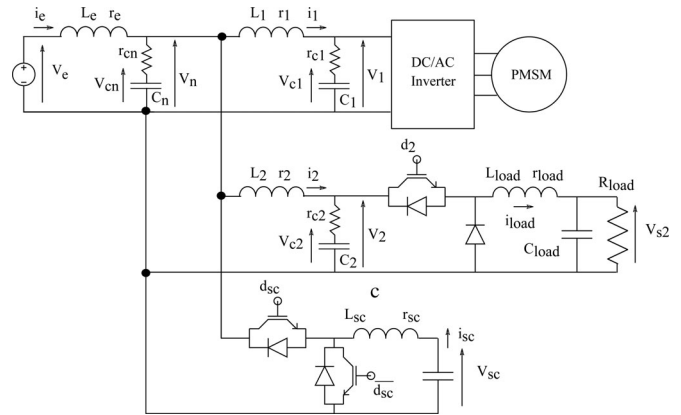


Fig. 1. Considered MVdc network.

nearby the different loads of the system and long cables may have to be used. This leads to inductive and resistive effects, which should be considered in the stability analysis. An inductor (L_e , r_e) is placed between the source and the capacitor C_n to model these parasitic effects. The capacitor C_n is defined as the main dc-bus of the network. Finally, for similar reasons but also to represent possible differential filters, inductors (L_1 , r_1 and L_2 , r_2) are placed between each load and the capacitors of the two secondary dc-bus (C_1 and C_2). Equivalent series resistors (ESR) of capacitances are also taken into account r_{cn} , r_{c1} , and r_{c2} . The network scheme is presented in Fig. 1.

B. Control System

A good control of the aforementioned MVdc network (see Fig. 1) permits optimization of the global efficiency of the system by a clever use of the SC. This can be realized by the implantation of global control laws, which organize and optimize the energy transfer in the network. In [5] and [6], multilevel controllers are used to control such a system. The lowest level is constituted by individual control of each load. The second level corresponds to a global supervisor that takes global information on the system state and produces references for the lowest control stage. In [6], a global supervisor based on the well-known fuzzy logic was proposed. Different objectives, such as energy-storage management, may be considered, and according to this, individual references are generated on the storage system. In [5], the authors used the flatness theory to generate an energy reference for the system components and combine it with a fuzzy logic algorithm to optimize the energy transfer on each ESD.

In general, the main role of ESDs in a dc-power network is to increase the global efficiency of a system by absorbing the energy, which can be provided by the actuators, and to compensate the transient mode after a high power demand from the loads. Other applications, such as active filtering or providing an energy source in the case of the main source failure, can also be considered.

Nevertheless, the aim of this paper is to show how a stabilizing law for dc power systems with adjustable effect on the energy management can be generated. In addition, it will be shown that the use of ESDs does not avoid instability. This study is made in two steps. We begin by considering only the first-level

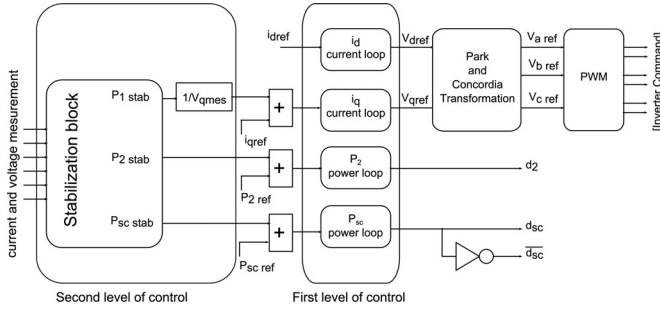


Fig. 2. Control system.

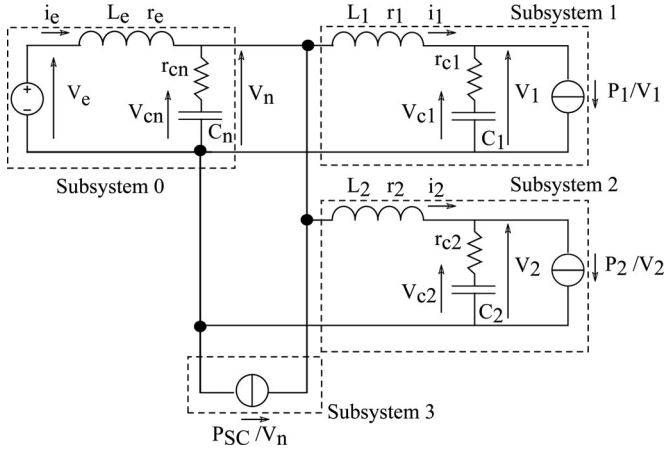


Fig. 3. Equivalent model of the electric system.

control: one control loop is implemented for the individual control of each load. Then, the stabilizing block is put on the second level as depicted in Fig. 2. Considering this, the two dc/dc converters are regulated with a power control loop as the PMSM is regulated with a torque (or current) control loop using a classical vector control [31].

To simplify the model, it is assumed that the current controllers are efficient so that the impact of variations on the dc-link voltage on the controlled currents is negligible if these latter are controllable. So, the inverter motor drive system can be modeled in a restrictive case from the stability point of view, i.e., a CPL. The control of the converters are also assumed to be efficient enough to consider the SC and its dc/dc converter as a perfect CPL. Considering this, we simplify the system shown in Fig. 1 by the one presented in Fig. 3. This simplified system does not permit study of a high-frequency phenomenon, but gives a good representation of the network, which is sufficiently precise to study the asymptotic stability of a dc-power network. These hypotheses will be confirmed in the next section by some experimental observations.

C. Experimental Bench

The system depicted in Fig. 1 was realized at the GREEN laboratory, Nancy, France. Control levels shown in Fig. 2 were implemented on a DSP for a real-time control of the system. Because the objective is to study the stability of a dc-power network, some parameters, particularly inductances, are chosen

TABLE I
EXPERIMENTAL SYSTEM PARAMETERS

$V_e = 200 \text{ V}$	$r_{c1} = 0.05 \Omega$
$r_e = 0.4 \Omega$	$r_2 = 0.42 \Omega$
$L_e = 17.3 \text{ mH}$	$L_2 = 19.6 \text{ mH}$
$C_n = 1050 \mu\text{F}$	$C_2 = 1050 \mu\text{F}$
$r_{cn} = 0.05 \Omega$	$r_{c2} = 0.05 \Omega$
$r_1 = 0.8 \Omega$	$C_{sc} = 165 \text{ F}$
$L_1 = 40 \text{ mH}$	$L_{sc} = 1 \text{ mH}$
$C_1 = 500 \mu\text{F}$	$L_2 = 19.6 \text{ mH}$
Switching frequency 10 kHz	
no. of pole pairs 4	
Stator resistance 0.5Ω	
Stator inductance 3.1 mH	
Torque constant 0.572 Nm/A	
Friction constant $14 \text{ g.m}^2/\text{s}$	
Inertia constant 3.7 g.m^2	

in such a manner that unstable oscillations appear at relatively low powers in order to avoid damaging the bench. Table I gives the different parameters of the experimental bench.

III. LOCAL ASYMPTOTIC STABILITY ANALYSIS

A. System Modeling

In the literature, several methods are available for considering the stability of nonlinear systems. They permit large-signal-stability analysis by the determination of an approximation of the domain of attraction of the operating point. In [16] and [27], the authors use the well-known Takagi–Sugeno multimodel representation for the nonlinear model and then use a linear matrix inequality (LMI) solver to obtain a Lyapunov function. This method has the advantage of being applicable to high-order systems and its efficacy depends on the LMI solver result. The less conservative the LMI resolution, the larger the domain of attraction estimation. The second method proposed in [17] and then in [20] is based on the Brayton and Mother theorems [32], [33]. These theorems construct a function that, under some conditions, can be considered as a Lyapunov function. In [34], the authors propose modification of the mixed-potential function to enlarge the estimated domain of attraction. Another approach is proposed in [18] and [19], where a genetic algorithm is applied to optimize the size of the domain of attraction obtained by using a proper Lyapunov function.

Other methods, based on the linearized system, are also available. In this case, the local stability of the operating point can be analyzed using poles of the linearized model. This is the most frequently used method for studying the stability of a system. In [7] and [9], a method has been proposed to study the stability of cascaded systems. The method is based on the linearization of the system around its operating point and the determination of input and output impedances of each electrical subsystems. Then, a voltage transfer function is developed and it is shown that the ratio of these impedances should verify certain conditions in order to prove the local stability. Graphical interpretation is possible using Nyquist and/or Bode diagrams, which allow consideration of graphical robustness margins.

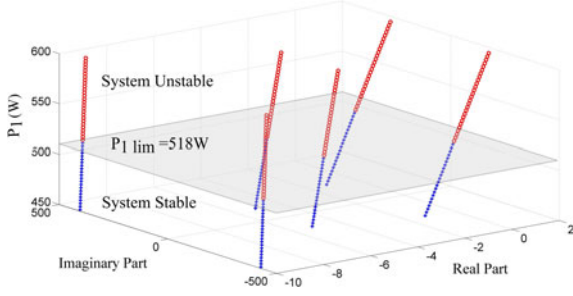


Fig. 4. Eigenvalues evolution.

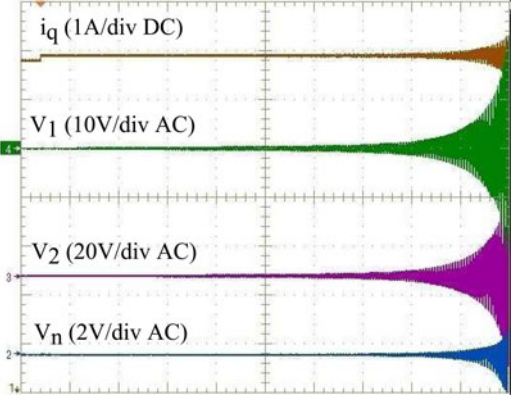
The dc-power network in Fig. 3 can be modeled to the form given in (1). We have a nonlinear system, where the nonlinearities are due to the CPLs. In this section, we will only study the local stability of the operating point of the system. This corresponds to studying the linearized model around its operating point. To do this, we consider the linearized input and output impedances of the different subsystems of the network (see Fig. 3)

$$\left\{ \begin{array}{l} L_e \frac{di_e}{dt}(t) = -r_e i_e(t) - r_{cn} C_n \frac{dV_{cn}}{dt} - V_{cn} + V_e(t) \\ L_1 \frac{di_1}{dt}(t) = -r_1 i_1(t) - r_{c1} C_1 \frac{dV_{c1}}{dt} - V_{c1} \\ \quad + r_{cn} C_n \frac{dV_{cn}}{dt} + V_{cn} \\ L_2 \frac{di_2}{dt}(t) = -r_2 i_2(t) - r_{c2} C_2 \frac{dV_{c2}}{dt} - V_{c2} \\ \quad + r_{cn} C_n \frac{dV_{cn}}{dt} + V_{cn} \\ C_n \frac{dV_{cn}}{dt}(t) = i_e(t) - i_1(t) - i_2(t) + \frac{P_{sc}}{r_{cn} C_n \frac{dV_{cn}}{dt} + V_{cn}} \\ C_1 \frac{dV_{c1}}{dt}(t) = i_1(t) - \frac{P_1}{r_{c1} C_1 \frac{dV_{c1}}{dt} + V_{c1}} \\ C_2 \frac{dV_{c2}}{dt}(t) = i_2(t) - \frac{P_2}{r_{c1} C_1 \frac{dV_{c1}}{dt} + V_{c1}} \end{array} \right. \quad (1)$$

B. Stability Study

To obtain I/O impedances of the subsystems shown in Fig. 3, each variable is expressed as the sum of its steady-state value and a small variation around it: $x(t) = x_0 + \tilde{x}(t)$. The voltage transfer function of the system is expressed by (2) according to the considered impedances presented in Fig. 3 and given in (3) and where $K(s)$ is a stable voltage transfer function. The local stability can be evaluated by studying the poles of (2). If there exists at least one unstable pole in $\tilde{V}_n/\tilde{V}_e$, then the system is unstable.

With the parameters given in Table I and for different values of P_1 , the poles of (2) are numerically determined and plotted in Fig. 4. This study fixes the stability limit at 518 W for the PMSM power with $P_2 = 350$ W, $P_{sc} = 0$ W, and $V_e = 200$ V. To confirm it, Fig. 5 shows measurements related to the system re-

Fig. 5. System voltage response after a load step from $i_q = 6.7$ A to $i_q = 6.8$ A.

sponse to a load step up to $P_1 = 545$ W. As can be seen, the voltage response gets slowly unstable confirming that we are at the limit between stable and unstable mode. As can be expected, the experimental stability limit is slightly different from the theoretical one as the approximation by CPLs corresponds to a restrictive stability case. This confirms that the model of Fig. 3, based on CPLs, is appropriate for the asymptotic stability analysis

$$\frac{\tilde{V}_n(s)}{\tilde{V}_e(s)} = \frac{K(s)}{1 + Z_{out0}(\frac{1}{Z_{in1}} + \frac{1}{Z_{in2}} + \frac{1}{Z_{in3}})} \quad (2)$$

$$\left\{ \begin{array}{l} Z_{out0}(s) = \frac{\tilde{V}_n(s)}{-\tilde{i}_n(s)} \Big|_{\tilde{V}_e(s)=0} = \frac{r_e + L_e s}{1 + C_n(r_{cn} + r_e)s + L_e C_n s^2} \\ Z_{in1}(s) = \frac{\tilde{V}_n(s)}{\tilde{i}_1(s)} = r_1 + L_1 s + \frac{1 + r_{c1} C_1 s}{C_1 s(1 - \frac{P_1 r_{c1}}{V_{10}^2}) - \frac{P_1}{V_{10}^2}} \\ Z_{in2}(s) = \frac{\tilde{V}_n(s)}{\tilde{i}_2(s)} = r_2 + L_2 s + \frac{1 + r_{c2} C_2 s}{C_2 s(1 - \frac{P_2 r_{c2}}{V_{20}^2}) - \frac{P_2}{V_{20}^2}} \\ Z_{in3}(s) = \frac{V_{n0}^2}{P_{sc}} \end{array} \right. \quad (3)$$

IV. ACTIVE STABILIZATION

A. Stabilization Block Implementation

As discussed earlier, the system in its original configuration becomes unstable if the power demand by the loads goes too high. To improve the stability margins, it is known that higher capacitances should be employed. But, this leads to increases in the size, weight, and cost of the system. In addition, the system parameters may vary due to temperature change or deterioration. This is the case with aluminum electrolytic capacitors, which are the leading choice for an ac variable-frequency drives bus filter, where the ESR may decrease down to 50% for a 100 °C variation of temperature [35]. Considering security margins is one way to handle this problem, but it can lead to a rise in the damping effect of the system by increasing the capacitor

size and/or the resistivity of the system. Because this type of solution is contrary to the need for reduced weight and size in transportation systems, we propose another way to improve the stability. The principle is to implement a centralized stabilizing supervisor on the whole system as suggested in Fig. 2. For practical and reliability reasons, decentralized supervisors are often more appropriate to industrial applications [36]. Their drawback lies in the lack of mathematical proof of the global stability of the whole system. This can be overcome when a centralized supervisor is used.

In order to stabilize the system, stabilizing powers ($P_{sc\text{stab}}$, $P_{1\text{stab}}$ and $P_{2\text{stab}}$) will be injected to each load. To study the stability of the operating point, it is usual to move it to the origin by introducing the new state variables in (5) and considering the new state vector given in (6). Then, the system represented in Fig. 3 can be modeled by the linear model (4) with A, B and U given respectively in (7)–(9).

$$\dot{X}(t) = AX(t) + BU(t) \quad (4)$$

$$\begin{cases} i_e(t) = i_{e0} + x_1(t) \\ i_1(t) = i_{10} + x_2(t) \\ i_2(t) = i_{20} + x_3(t) \\ V_{cn}(t) = V_{n0} + x_4(t) \\ V_{c1}(t) = V_{10} + x_5(t) \\ V_{c2}(t) = V_{20} + x_6(t) \\ V_n(t) = V_{n0} + x_7(t) \\ V_1(t) = V_{10} + x_8(t) \\ V_2(t) = V_{20} + x_9(t) \end{cases} \quad (5)$$

$$X = [x_1 \ x_2 \ x_3 \ x_4 \ x_5 \ x_6]^t \quad (6)$$

$$A = \begin{bmatrix} \frac{-(r_e + r_{cn})}{L_e} & \frac{r_{cn}}{L_e} & \frac{r_{cn}}{L_e} \\ \frac{r_{cn}}{L_1} & \frac{-(r_1 + r_{cn} + r_{c1})}{L_1} & \frac{-r_{cn}}{L_1} \\ \frac{r_{cn}}{L_2} & \frac{-r_{cn}}{L_2} & \frac{-(r_2 + r_{cn} + r_{c2})}{L_2} \\ \frac{1}{C_n} & \frac{-1}{C_n} & \frac{-1}{C_n} \\ 0 & \frac{1}{C_1} & 0 \\ 0 & 0 & \frac{1}{C_2} \\ \frac{-1}{L_e} & 0 & 0 \\ \frac{1}{L_1} & \frac{-1}{L_1} & 0 \\ \frac{1}{L_2} & 0 & \frac{-1}{L_2} \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad (7)$$

$$B = \begin{bmatrix} \frac{r_{cn}}{L_e} & 0 & 0 \\ \frac{-r_{cn}}{L_1} & \frac{-r_{c1}}{L_1} & 0 \\ \frac{-r_{cn}}{L_2} & 0 & \frac{-r_{c2}}{L_2} \\ \frac{-1}{C_n} & 0 & 0 \\ 0 & \frac{1}{C_1} & 0 \\ 0 & 0 & \frac{1}{C_2} \end{bmatrix} \quad (8)$$

$$U(t) = \begin{bmatrix} u_1(t) \\ u_2(t) \\ u_3(t) \end{bmatrix} = \begin{bmatrix} \frac{P_{sc}(t)x_7(t) - P_{sc\text{stab}}(t)V_{n0}}{V_{n0}V_n(t)} \\ \frac{P_1x_8(t) - P_{1\text{stab}}(t)V_{10}}{V_{10}V_1(t)} \\ \frac{P_2x_9(t) - P_{2\text{stab}}(t)V_{20}}{V_{20}V_2(t)} \end{bmatrix} \quad (9)$$

From (9), we obtain a relation expressing stabilizing powers, which should be added to the reference powers of the loads. In order to stabilize the system and generate $P_{sc\text{stab}}$, $P_{1\text{stab}}$ and $P_{2\text{stab}}$, we design U in such a way to ensure system global asymptotic stability. For this, we choose the feedback form (10) with D as a diagonal positive-definite matrix and P as a solution of the Lyapunov equation (11), where P and Q are the positive-definite symmetric matrices [37]

$$U = -(BD)^tPX \quad (10)$$

$$PA + A^tP = -Q. \quad (11)$$

With this feedback, (4) is modified into

$$\dot{X}(t) = (A - B(BD)^tP)X(t). \quad (12)$$

This feedback law ensures that the matrix $(A - B(BD)^tP)$ is a solution of the Lyapunov equation, which proves the global stability of the system.

Proof: $P(A - B(BD)^tP) + (A - B(BD)^tP)^tP = PA + A^tP - PB(BD)^tP - (B(BD)^tP)^tP$ with $PA + A^tP < 0$, so it is sufficient to verify that $PB(BD)^tP + (B(BD)^tP)^tP > 0$.

As P and D are both positive definite, we can put them in the following form: $P = M_P M_P^t = M_P^t M_P$ and $D = M_D M_D^t = M_D^t M_D$ with M_D and M_P being the invertible matrices.

So, $PB(BD)^tP + (B(BD)^tP)^tP = (M_P M_P^t B M_D)(M_P M_P^t B M_D)^t + (P^t B M_D)(P^t B M_D)^t > 0$.

Remark: This proof is independent of the loads power and the system operating point.

Finally, the stabilizing powers are given by (13) according to (9). The role of the matrix D is to place some coefficients on each stabilization output in order to adjust the contribution of each load, in particular ESDs (the SC here), on the stabilization of the whole system. This drastically reduces, if necessary, the

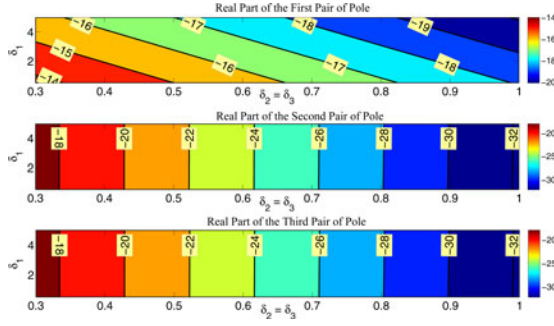


Fig. 6. Evolution of the real part of pole pairs of the stabilized system.

contribution of the loads

$$\begin{cases} P_{sc\text{stab}}(t) = (i_{10} + i_{20} - i_{e0})x_7(t) - u_1(t)V_n(t) \\ P_{1\text{stab}}(t) = i_{10}x_8(t) - u_2(t)V_1(t) \\ P_{2\text{stab}}(t) = i_{20}x_9(t) - u_3(t)V_2(t) \end{cases} \quad (13)$$

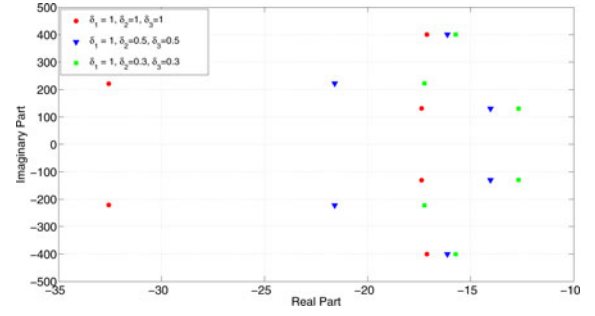
$$D = \begin{bmatrix} \delta_1 & 0 & 0 \\ 0 & \ddots & 0 \\ 0 & 0 & \delta_n \end{bmatrix} \quad \delta_i > 0 \quad \forall i \in [1, n]. \quad (14)$$

Eigenvalues of the matrix $(A - B(BD)^t P)$ give the dynamic of the stabilized system and permit tuning of the stabilization blocks. The closer the real part of the eigenvalues is to the semipositive half-plane, the less the system is damped and the less important the stabilizing power is. With such a tuning, the loads power will be modified in a negligible manner, but the system transient response time may be long in this case. On the other hand, fast transient dynamics imply important stabilizing powers to be provided by the loads. This may significantly disturb normal operating of the loads and may be undesirable. So, a good tradeoff would be that each stabilizing power in (13) does not exceed a certain limit (in percentage of its power demand) during the transient mode.

To fix these limits, we consider the system response time constant τ . We fix a time interval $[\tau_{\min}, \tau_{\max}]$, where $\tau_{\max}$ is the maximum time constant for the stabilized system and $\tau_{\min}$ is the minimum time constant that we will allow. By limiting $\tau_{\min}$, we avoid too high dynamics during transients and so limit the stabilizing powers. This is equivalent to limiting the real part of the closed-loop system eigenvalues in the interval I_{dyn} defined by (15)

$$I_{\text{dyn}} = [-1/\tau_{\min}, -1/\tau_{\max}]. \quad (15)$$

In order to take this into consideration, matrix Q is chosen equal to $10^{-5}I_6$ with I_n being the n th-order identity matrix. The real part values of the matrix $(A - B(BD)^t P)$ eigenvalues for different values of the coefficients in D are studied in Fig. 6. The real part values are indicated by the given color bar. It indicates the system dynamic variation according to the choice of δ_1 , δ_2 , and δ_3 . In order to decrease the stabilization impact on the load references, coefficient δ_1 , which corresponds to the SC, is raised, whereas δ_2 and δ_3 are reduced. Fig. 6 represents the evolution of the real part of the three pairs of poles of the system with respect to D . On the abscissa, both δ_2 and δ_3 are moving

Fig. 7. Stabilized system eigenvalues for different values of D .

together and δ_1 is moving on the ordinate axis. It is shown that, with a simultaneous decrease of δ_2 and δ_3 , the time constant of the transient mode increases. This is consistent with the fact that the less these coefficients are, the less important the stabilization powers are and the less damped the system is. The evolution of δ_1 does not significantly affect the real part of the eigenvalues. This can be justified by the fact that the SC is situated on a less critical point of the network in comparison with the two other loads, which are connected to the main dc-bus through poorly damped filters. The evolution of the eigenvalues confirms the role of the matrix D on the tuning of the stabilization efforts. The matrix D proposes that the designer choose and weight the role of each load of the system on its stabilization. The most sensitive loads, like actuators, can be protected, whereas ESDs can contribute to the stabilization of the system.

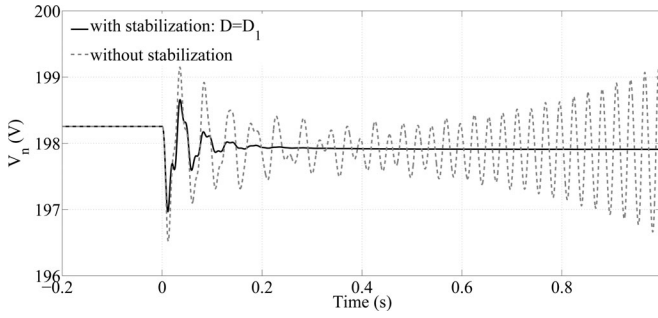
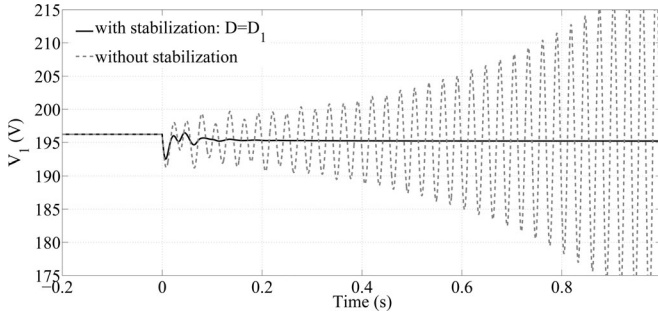
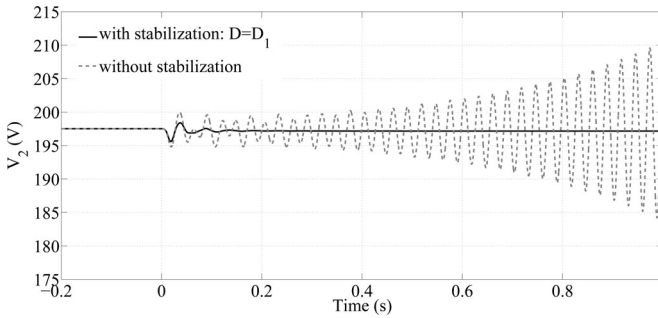
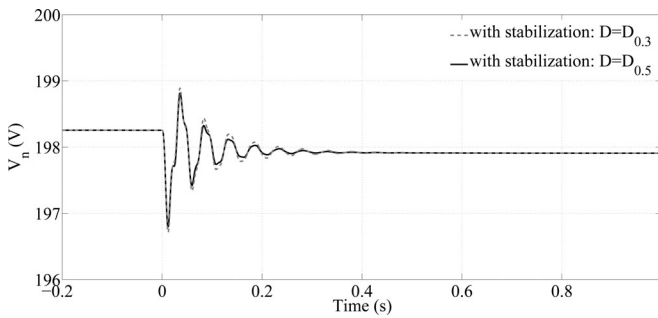
B. Simulation Results

To validate the principle of the proposed method, a model of the system represented in Fig. 1 is developed on MATLAB-Simulink. A d - q model is used for the PMSM. The simulation parameters are those given in Table I and the following matrices D are tested

$$D_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad D_{0.5} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0.5 & 0 \\ 0 & 0 & 0.5 \end{bmatrix}, \quad (16)$$

$$D_{0.3} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0.3 & 0 \\ 0 & 0 & 0.3 \end{bmatrix}.$$

Fig. 7 shows the stabilized system eigenvalues for the three aforementioned matrices D . It confirms the observations in Fig. 6. The less the coefficients δ_1 and δ_2 are, the less the damping effect on the state variables of the electrical system is. This is confirmed by the simulation results given in Figs. 8–13. The system, without the active stabilization, is unstable for the considered load step, whereas it is stable with the proposed stabilization method. We can also see that the system voltage and current dynamics verify the poles map given in Fig. 7. Next, the experimental results corresponding to these simulation tests are presented.

Fig. 8. Response of V_n after a i_q step from 6.6 to 7.6 A (simulation).Fig. 9. Response of V_1 after a i_q step from 6.6 to 7.6 A (simulation).Fig. 10. Response of V_2 after a i_q step from 6.6 to 7.6 A (simulation).Fig. 11. Response of V_n after a i_q step from 6.6 to 7.6 A (simulation).

C. Fault-Tolerance and Operating in Degraded Modes

The use of a centralized structure may be criticized as it creates a single point whose failure is harmful for the system. In aircraft industry, to improve the reliability, it is common to use the parallel redundancy by multiplying the number of subsystems. So, the use of multiple sensors for the same measurement

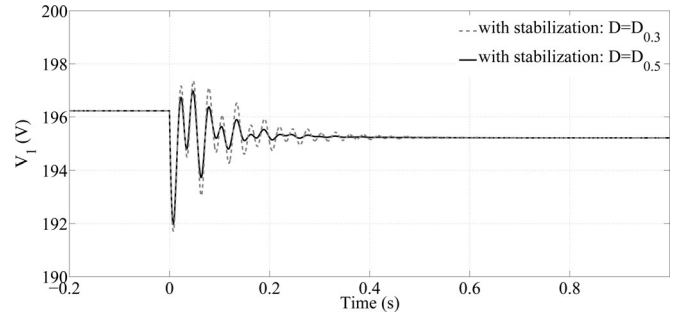
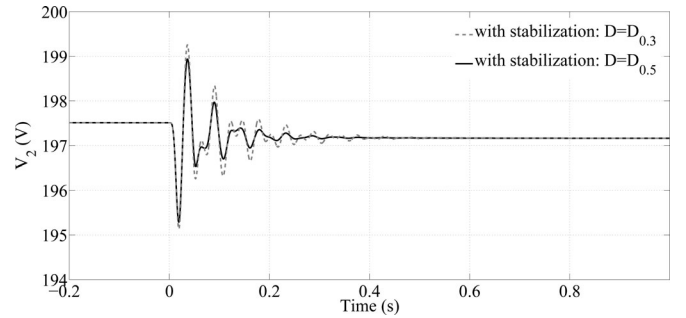
Fig. 12. Response of V_1 after a i_q step from 6.6 to 7.6 A (simulation).Fig. 13. Response of V_2 after a i_q step from 6.6 to 7.6 A (simulation).

TABLE II
STABILIZATION BLOCK FAILURE CASES, $D = D_1$

Failure case	$P_{1\text{ stab}} = 0$	$P_{2\text{ stab}} = 0$	$P_{sc\text{ stab}} = 0$
$P_{1\text{ lim}}$ (W)	607	3000	3000
$P_{2\text{ lim}}$ (W)	3000	1952	3000
$P_{sc\text{ lim}}$ (W)	± 3000	± 3000	± 1485

is frequent. In this case, we can assume that all the measurements (voltages, currents) are available in the central stabilization block.

Another failure case may occur when a stabilizing signal is not transmitted to its corresponding load. In this degraded mode, the stabilization block can no more ensure the global stability of the system as the proof given earlier is no more applicable. Nevertheless, the other stabilizing signals impose still stabilizing feedbacks on their respective loads, which may permit to damp the system response. In this case, by using a small signal model of the degraded system and applying a conventional linear analysis tool, we can determine the limit on the loads power for which the system local asymptotic stability is still ensured. These limits depend of course on the operating point and on the faulty load. Here, we choose 3 kW for the loads 1 and 2 and ± 3 kW for the SC. Then, the following power limits are obtained when one of the stabilizing signals misses (see Table II). If, for each case, the absorbed power by each load satisfies its corresponding limit, the local asymptotic stability of the system is still preserved.

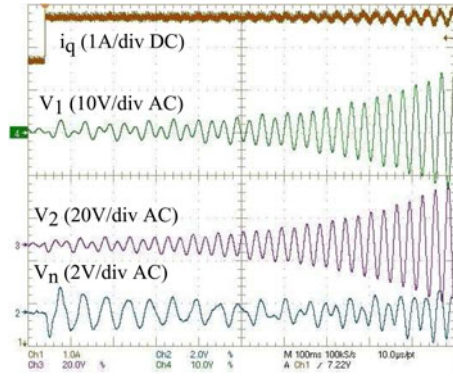


Fig. 14. V_n , V_1 , and V_2 experimental response after a i_q step from 6.6 to 7.6 A without stabilization block.

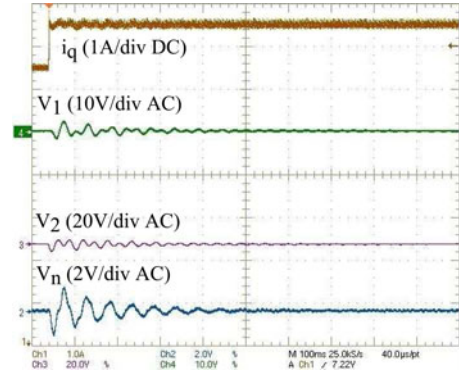


Fig. 16. V_n , V_1 , and V_2 experimental response after a i_q step from 6.6 to 7.6 A with $\delta_1 = 1$, $\delta_2 = 0.5$, and $\delta_3 = 0.5$.

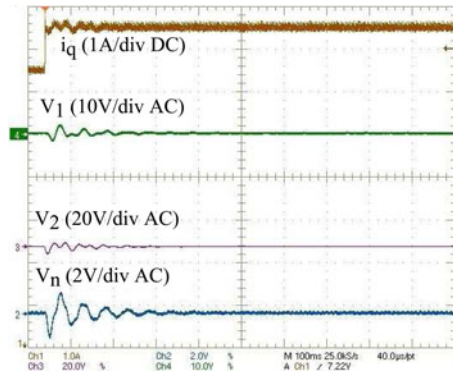


Fig. 15. V_n , V_1 , and V_2 experimental response after a i_q step from 6.6 to 7.6 A with $\delta_1 = 1$, $\delta_2 = 1$, and $\delta_3 = 1$.

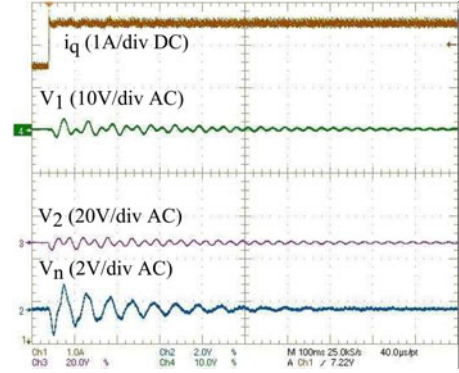


Fig. 17. V_n , V_1 , and V_2 experimental response after a i_q step from 6.6 to 7.6 A with $\delta_1 = 1$, $\delta_2 = 0.3$ and $\delta_3 = 0.3$.

V. EXPERIMENTAL RESULTS

As was presented in Section II, an experimental bench was developed at the GREEN Laboratory for studying the stability of an isolated dc-power network. It allowed validation of the small-signal-stability analysis and the CPL hypothesis in Section II (see Fig. 5). In the following, we perform experimental tests for evaluating the efficacy of the proposed stabilization method. The latter is implemented in the digital control card, which generates the stabilizing powers and sends them to the corresponding loads controllers. As described in the previous section, these powers depend on the tuning parameters δ_1 , δ_2 , and δ_3 . The presented results are measured by sufficiently high bandwidth sensors (>150 kHz) and sampled either by a 20-MHz oscilloscope (voltages) or by the digital control card (other variables).

The experimental results presented in Figs. 14–17, show the efficacy of the proposed active stabilization method and confirm the simulation results on the voltages and the currents dynamics. Indeed, the system is stabilized at an operating point, which would otherwise be unstable.

In Figs. 18–26, we can see the measured powers P_{sc} and P_2 and the i_q current of the PMSM. After a torque step on the PMSM, the stabilizing powers are injected into each load power reference, which disturb their usual operating. These disturbances are shown in Figs. 18–26. To compare different cases, a numerical indicator is defined in (17), where N is the number of sampled points. It is applied to each stabilizing power

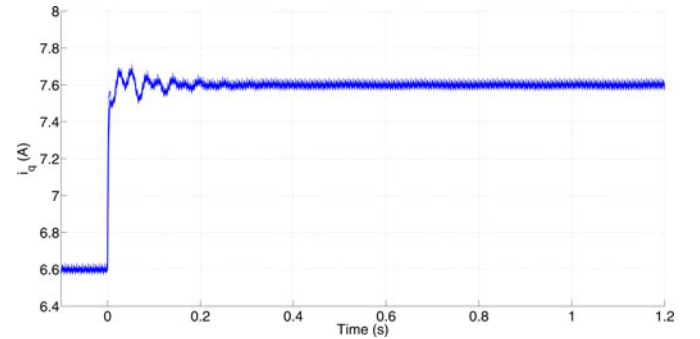


Fig. 18. Experimental q -axis current i_q after a i_q step from 6.6 to 7.6 A with $\delta_1 = 1$, $\delta_2 = 1$ and $\delta_3 = 1$.

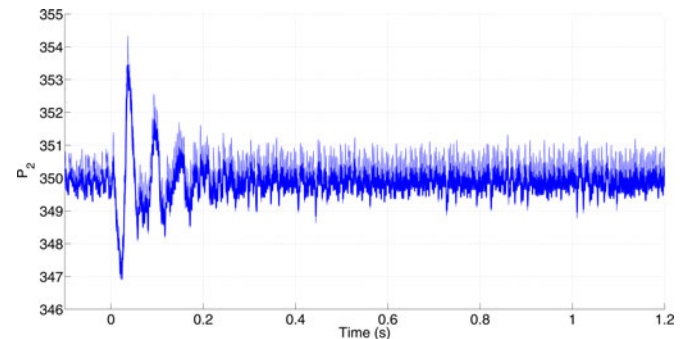


Fig. 19. Experimental load 2 power ($P_2 = P_{20} + P_{2\text{stab}}$) with $P_{20} = 350$ W after a i_q step from 6.6 to 7.6 A with $\delta_1 = 1$, $\delta_2 = 1$ and $\delta_3 = 1$.

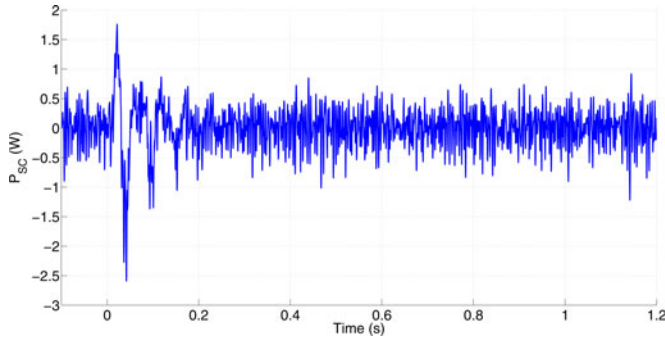


Fig. 20. Experimental SC power ($P_{SC} = P_{SC0} + P_{sc\text{stab}}$) with $P_{SC0} = 0$ W after a i_q step from 6.6 to 7.6 A with $\delta_1 = 1$, $\delta_2 = 1$ and $\delta_3 = 1$.

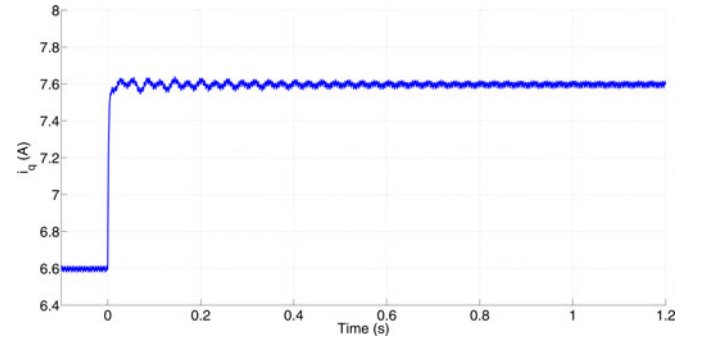


Fig. 24. Experimental q-axis current i_q after a i_q step from 6.6 to 7.6 A with $\delta_1 = 1$, $\delta_2 = 0.3$, and $\delta_3 = 0.3$.

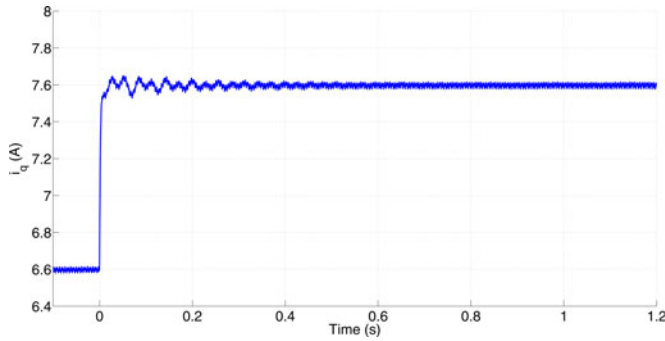


Fig. 21. Experimental q-axis current i_q after a i_q step from 6.6 to 7.6 A with $\delta_1 = 1$, $\delta_2 = 0.5$, and $\delta_3 = 0.5$.

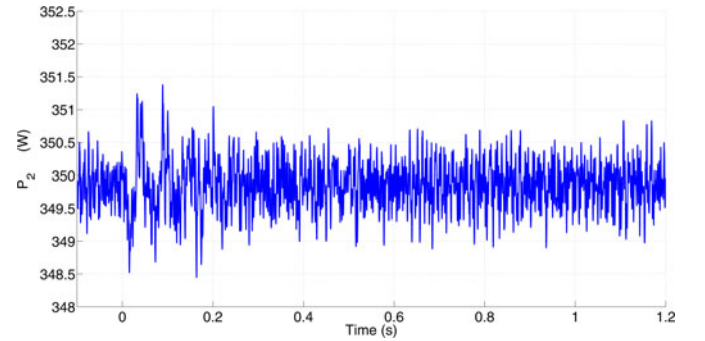


Fig. 25. Experimental load 2 power ($P_2 = P_{20} + P_{2\text{stab}}$) with $P_{20} = 350$ W after a i_q step from 6.6 to 7.6 A with $\delta_1 = 1$, $\delta_2 = 0.3$, and $\delta_3 = 0.3$.

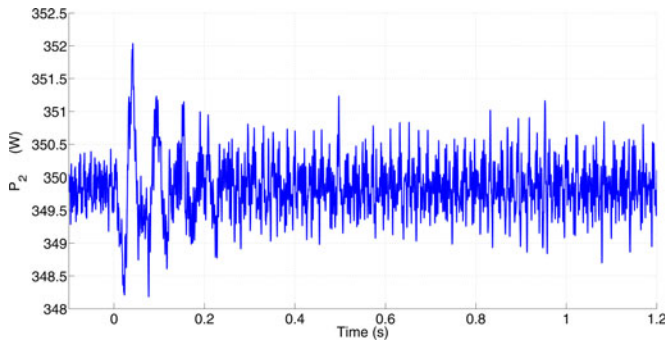


Fig. 22. Experimental load 2 power ($P_2 = P_{20} + P_{2\text{stab}}$) with $P_{20} = 350$ W after a i_q step from 6.6 to 7.6 A with $\delta_1 = 1$, $\delta_2 = 0.5$, and $\delta_3 = 0.5$.

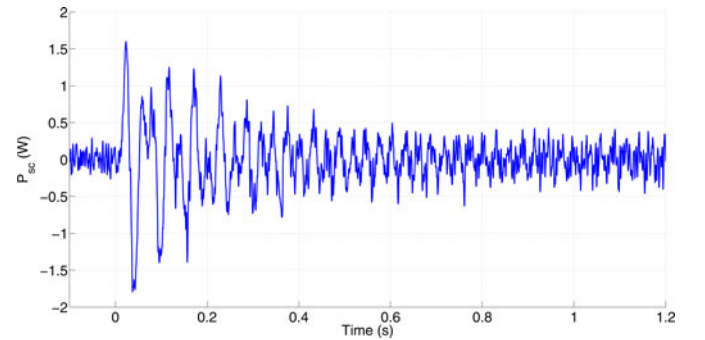


Fig. 26. Experimental SC power ($P_{SC} = P_{SC0} + P_{sc\text{stab}}$) with $P_{SC0} = 0$ W after a i_q step from 6.6 to 7.6 A with $\delta_1 = 1$, $\delta_2 = 0.3$, and $\delta_3 = 0.3$.

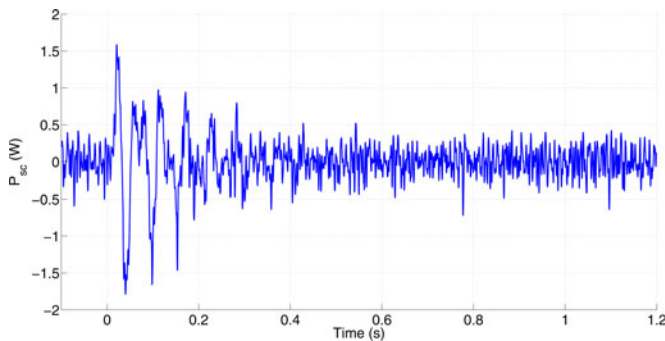


Fig. 23. Experimental SC power ($P_{SC} = P_{SC0} + P_{sc\text{stab}}$) with $P_{SC0} = 0$ W after a i_q step from 6.6 to 7.6 A with $\delta_1 = 1$, $\delta_2 = 0.5$, and $\delta_3 = 0.5$.

for the three cases. The results, given in Table III, confirm the impact of the matrix D on the stabilizing efforts. For example, $\langle P \rangle_{1\text{stab}}$ decreases respectively by 27% or 38% if $D_{0.5}$ or $D_{0.3}$ is used instead of D_1 . At the same time, $\langle P \rangle_{sc\text{stab}}$ increases respectively by 33% or 70%

$$\langle P \rangle = \sum_{k=1}^N \sqrt{\frac{P_k^2}{N}}. \quad (17)$$

This impact is also visible in the experimental results. In Figs. 18 and 19 with the matrix D_1 , the amplitude of p_2 varies between 347 and 354 W and that of the i_q current from 7.48 to 7.68 A. These variations correspond to, respectively, 2% and 2.6% of the dc/dc power reference ($=350$ W) and the i_q PMSM drive reference ($=7.6$ A). With the matrix $D_{0.3}$, these

TABLE III
EXPERIMENTAL STABILIZING POWERS

	D_1	$D_{0.5}$	$D_{0.3}$
$< P >_1 \text{ stab}$	1.108	0.808	0.689
$< P >_2 \text{ stab}$	0.451	0.300	0.229
$< P >_{sc} \text{ stab}$	0.194	0.259	0.329

variations are, respectively, limited to 0.8% and 0.4% of the references values. This shows that, with a good choice of D , we can ensure system stability with a negligible impact on the loads.

These experimental results, as well as those obtained by simulation, show that the required energy for stabilizing the system is not significant. In a configuration where the designers want to install an ESD with the system stability as the only objective, a simple capacitor would be sufficient and there is no need to add the SC. Meanwhile, the latter is often employed in the network for energy-management purposes and could be used for stabilizing the system if a proper stabilizer is implemented.

VI. CONCLUSION

This paper has presented a method for implementing a global stabilization block on a dc-power network. At first, a small signal stability study of the system is proposed using the I/O impedances of its subsystems. Loads are modeled as CPLs, which allows a simple representation. This is sufficiently precise for stability considerations. It permits local stability of the linearized system around its operating point. The small signal stability analysis underlines that, in the conception of dc-power networks and their use, stability issues must be taken into consideration in the determination of the power limits that the loads have to verify. Otherwise, the system risks unstable oscillations, which cause system failures or even damage the network components. These limits depend, of course, on capacitances used in the system.

To avoid these limits and/or reduce the capacitor size, and thus the weight and the cost of the system, we propose a stabilization method ensuring global stability. This method permits stabilizing powers that, added to each load power reference, avoid the system instability. These stabilizing powers are generated using a Lyapunov equation, which is verified for the whole system. The proposed method does not depend on the type of loads. Its implementation depends only on the network structure. The method also permits the adjustment of the contribution of each load in the system stabilization. Their corresponding stabilizing powers can be decreased in a certain way to reduce the stabilization impact on them. The simulation and experimental results confirm the efficacy of the proposed method.

Nevertheless, even if this method ensures the global stability of the system, it imposes upon the designer some technical constraints, due to the important number of required sensors. As it is a centralized method, all the data have to be transmitted to the central stabilization block, and then, all the stabilizing powers have to be sent to each load control block. The bandwidth of the

communication links should be sufficient to refresh the data for each sampling period. In addition, these data buses may also lead to electromagnetic compatibility problems. Some existing standards, such as MIL-STD-1553 and AFDX, can be used.

In summary, the main contribution of this work is the mathematical development and experimental validation of a global stabilizing method for dc-power systems.

REFERENCES

- [1] L. Romeral, J. Rosero, A. Espinosa, J. Cusido, and J. Ortega, "Electrical monitoring for fault detection in an EMA," *Aerosp. Electron. Syst. Mag.*, vol. 25, no. 3, pp. 4–9, Mar. 2010.
- [2] N. Velly, N. Takorabet, F. Meibody-Tabar, P. Liegeois, F. Nierlich, F. Leynaert, and G. Humbert, "Double channel pm motor for avionic applications: Impact of winding topologies," in *Proc. IEEE Energy Conver. Congr. Expo.*, 2009, pp. 2387–2394.
- [3] F. Daigle, I. Roychoudhury, G. Biswas, X. Koutsoukos, A. Patterson-Hine, and S. Poll, "A comprehensive diagnosis methodology for complex hybrid systems: A case study on spacecraft power distribution systems," *IEEE Trans. Syst., Man, Cybern. A, Syst. Humans*, vol. 40, no. 5, pp. 917–931, Sep. 2010.
- [4] N. Takorabet, J. Caron, B. Vaseghi, B. Nahid-Mobarakeh, F. Meibody-Tabar, and G. Humbert, "Study of different architectures of fault tolerant actuator using double star PM motor," in *Proc. Ind. Appl. Soc.*, 2008, pp. 1–6.
- [5] M. Zandi, A. Payman, J. Martin, S. Pierfederici, B. Davat, and F. Meibody-Tabar, "Flatness based control of a hybrid power source with fuel cell/supercapacitor/battery," in *Proc. IEEE Energy Conver. Congr. Expo.*, 2010, pp. 1629–1634.
- [6] H. Zhang, F. Mollet, C. Saudemont, and B. Robyns, "Experimental validation of energy storage system management strategies for a local dc distribution system of more electric aircraft," *IEEE Trans. Ind. Electron.*, vol. 57, no. 12, pp. 3905–3916, Dec. 2010.
- [7] R. Middlebrook, "Input filter consideration in design and application of switching regulators," in *Proc. Ind. Appl. Soc.*, 1976, pp. 366–382.
- [8] A. Emadi, B. Fahimi, and M. Ehsani, "On the concept of negative impedance instability in the more electric aircraft power systems with constant power loads," SAE Technical Paper 1999-01-2545, 1999.
- [9] R. Middlebrook, "Design techniques for preventing input-filter oscillations in switched-mode regulators," in *Proc. POWERCON*, 1978, pp. 153–168.
- [10] S. Sudhoff, K. Corzine, S. Glover, H. Hegner, J. Robey, and H.N., "DC-link stabilized field oriented control of electric propulsion systems," *IEEE Trans. Energy Convers.*, vol. 13, no. 1, pp. 27–33, Mar. 1998.
- [11] P. Liutanakul, S. Pierfederici, and F. Meibody-Tabar, "DC-link capacitor reduction of a controlled rectifier supplying n inverter-motor drive systems by compensating the load variations," in *Proc. IEEE Power Electron. Spec. Conf.*, 2004, vol. 2, pp. 1298–1303.
- [12] A. Emadi, A. Khaligh, C. Rivetta, and G. Williamson, "Constant power loads and negative impedance instability in automotive systems: definition, modeling, stability, and control of power electronic converters and motor drives," *IEEE Trans. Veh. Technol.*, vol. 55, no. 4, pp. 1112–1125, Jul. 2006.
- [13] H. Mosskull, J. Galic, and B. Wahlberg, "Stabilization of induction motor drives with poorly damped input filters," *IEEE Trans. Ind. Electron.*, vol. 54, no. 5, pp. 2724–2734, Aug. 2007.
- [14] P. Liutanakul, A. Awan, S. Pierfederici, B. Nahid-Mobarakeh, and F. Meibody-Tabar, "Linear stabilization of a dc bus supplying a constant power load: A general design approach," *IEEE Trans. Power Electron.*, vol. 25, no. 2, pp. 475–488, Feb. 2010.
- [15] M.-A. Shamsi-Nejad, B. Nahid-Mobarakeh, S. Pierfederici, and F. Meibody-Tabar, "Fault tolerant and minimum loss control of double-star synchronous machines under open phase conditions," *IEEE Trans. Ind. Electron.*, vol. 55, no. 5, pp. 1956–1965, May 2008.
- [16] D. Marx, S. Pierfederici, B. Nahid-Mobarakeh, and B. Davat, "Contribution to determination of domain of attraction in power systems: Application to drives with input filter," in *Proc. Ind. Appl. Soc.*, 2009, pp. 1–8.
- [17] M. Belkhat, R. Cooley, and A. Witulski, "Large signal stability criteria for distributed systems with constant power loads," in *Proc. IEEE Power Electron. Spec. Conf.*, 1995, vol. 2, pp. 1333–1338.

- [18] B. Loop, S. Sudhoff, and Z. S. H., E. Zivi, "An optimization approach to estimating stability regions using genetic algorithms," in *Proc. Amer. Control Conf.*, 2005, vol. 1, pp. 231–236.
- [19] B. Loop, D. Sudhoff, S. Zak, and L. Zivi, "Estimating regions of asymptotic stability of power electronics systems using genetic algorithms," *IEEE Trans. Control Syst. Technol.*, vol. 18, no. 5, pp. 1011–1022, Sep. 2010.
- [20] A. Griffo, J. Wang, and D. Howe, "Large signal stability analysis of dc power systems with constant power loads," in *Proc. Vehicle Power Propulsion Conf.*, 2008, pp. 1–6.
- [21] M. Cespedes, T. Beechner, L. Xing, and J. Sun, "Stabilization of constant-power loads by passive impedance damping," in *Proc. 25th Annu. IEEE Appl. Power Electron. Conf. Expo.*, 2010, pp. 2174–2180.
- [22] L. Harnefors and K. Pietläinenardier, "Inverter dc-link stabilizing control with improved voltage sag ride-through capability," in *Proc. Eur. Power Electron. Appl.*, 2005, pp. 1–10.
- [23] W. Lee and S. Sul, "DC-link voltage stabilization for reduced dc-link capacitor inverter," in *Proc. Energy Conver. Congr. Expo.*, 2009, pp. 1740–1744.
- [24] X. Liu, A. Forsyth, and A. Cross, "Negative input-resistance compensator for a constant power load," *IEEE Trans. Ind. Electron.*, vol. 54, no. 6, pp. 3188–3196, Dec. 2007.
- [25] S. Awan, A. B. Pierfederici, B. Nahid-Mobarakeh, and F. Meibody-Tabar, "Active stabilization of a poorly damped input filter supplying a constant power load," in *Proc. Energy Conver. Congr. Expo.*, 2009, pp. 2991–2997.
- [26] S. Awan, A. B. Pierfederici, B. Nahid-Mobarakeh, and F. Meibody-Tabar, "Stabilization of a dc-bus supplying a constant power load," in *Proc. Ind. Appl. Soc.*, 2009, pp. 1–8.
- [27] P. Magne, S. Pierfederici, and B. Nahid-Mobarakeh, "DC-link voltage large signal stabilization and transient control using a virtual capacitor," in *Proc. Ind. Appl. Soc.*, 2010, pp. 1–8.
- [28] S. Sanders and G. Verghese, "Lyapunov-based control for switched power converters," *IEEE Trans. Power Electron.*, vol. 7, no. 1, pp. 17–24, Jan. 1992.
- [29] V. Arcidiacono, A. Monti, and G. Sulligoi, "An innovative generation control system for improving design and stability of shipboard medium-voltage dc integrated power system," in *Proc. IEEE Electr. Ship Technol. Symp.*, Apr. 2009, pp. 152–156.
- [30] X. Liu, N. Fournier, and A. Forsyth, "Active stabilisation of an HVdc distribution system with multiple constant power loads," in *Proc. Vehicle Power Propulsion Conf.*, 2008, pp. 1–6.
- [31] B. Bose, *Power Electronics and Motor Drives: Advances and Trends*. New York: Academic, 2006.
- [32] R. Brayton and J. Moser, "A theory of nonlinear networks—Part I," *Q. Appl. Math.*, vol. XXII, pp. 1–33, Apr. 1964.
- [33] R. Brayton and J. Moser, "A theory of nonlinear networks—Part II," *Q. Appl. Math.*, vol. XXII, pp. 81–104, Jul. 1964.
- [34] R. Ortega, D. Jeltsema, and J. Scherpen, "Power shaping: A new paradigm for stabilization of nonlinear RLC circuits," *IEEE Trans. Autom. Control*, vol. 48, no. 10, pp. 1762–1767, Oct. 2003.
- [35] L. Gasperi, "Life prediction modeling of bus capacitors in ac variable-frequency drives," *IEEE Trans. Ind. Appl.*, vol. 41, no. 6, pp. 1430–1435, Nov. 2010.
- [36] S. McArthur, E. Davidson, V. Catterson, A. Dimeas, N. Hatziaargyriou, F. Ponci, and T. Funabashi, "Multiagent systems for power engineering applications—Part I: Concepts, approaches, and technical challenges," *IEEE Trans. Power Syst.*, vol. 22, no. 4, pp. 1743–1752, Nov. 2007.
- [37] H. Khalil, *Nonlinear Systems*, 3rd ed. Englewood Cliffs, NJ: Prentice-Hall, 2002.



Pierre Magne (S'11) received the M.Sc. degree in electrical and electronics engineering from the Institut National Polytechnique de Lorraine (INPL), Nancy, France, in 2009, and the M.Eng. degree in electrical engineering from Ecole Nationale Supérieure d'Electricité et de Mécanique (ENSEM), Nancy, France, in 2009. He is currently working toward the Ph.D. degree in electrical engineering at the University of Lorraine, France.

His current research interests include power electronics, nonlinear control, and their applications.



Babak Nahid-Mobarakeh (M'05) received the B.Sc. degree from K.N. Toosi University of Technology, Tehran, Iran in 1991, the M.Sc. degree from the University of Tehran, Tehran, in 1995, and the Ph.D. degree from the Institut National Polytechnique de Lorraine (INPL), Nancy, France, in 2001, all in electrical engineering.

From 2001 to 2006, he was an Assistant Professor at the Centre de Robotique, Electrotechnique et Automatique (CREA), University of Picardie, Amiens, France. Currently, he is with Groupe de Recherche en Electronique et en Electrotechnique de Nancy (GREEN), Ecole Nationale Supérieure d'Electricité et de Mécanique of Nancy, INPL, Nancy, France. His main research interests include nonlinear and robust control techniques applied to the power systems.



Serge Pierfederici received the Engineer degree from Ecole Nationale Supérieure d'Electricité et de Mécanique (ENSEM), Nancy, France, in 1994 and the Ph.D. degree from the Institut National Polytechnique de Lorraine (INPL), Nancy, France, in 1998.

Since 1999, he has been with the INPL, where he is currently a Full Professor. His research activities in Groupe de Recherche en Electronique et en Electrotechnique de Nancy include the stability study of distributed power system and the control of multisources-multiload systems.